



Morphological and structural properties of isotactic polypropylene filled with nano-zinc oxide as investigated by dynamic rheology, creep and recovery in shear

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Abstract

Morphological and structural properties of nano-zinc oxide/isotactic polypropylene (nano-ZnO/iPP) composites with different nano-ZnO concentrations before and after compatibilization were investigated by scanning electron microscopy (SEM), dynamic rheology, shear creep and creep-recovery measurements. The complex viscosity and storage modulus of the composite melts first decrease and then increase with increasing nano-ZnO loading. The pseudo-solid-like behavior appearing in high-loaded composites could be attributed to the formation of the filler percolation network resulting from particle–particle interactions. The incorporating of maleic anhydride/styrene-grafted random copolypropylene (MPP) brings about the reduction of the number of big agglomerates of nano-ZnO, followed by the decrease in modulus and viscosity. Large nano-ZnO agglomerates or more complete percolation network formation enhances the creep resistance of iPP. This finding is explained by the existence of long relaxation time in the high-loaded system assigned to the relaxation of the iPP chains attached to the particle surface. Moreover, MPP has a pronounced influence on the shear creep and recovery behavior of the composites. Weighted relaxation spectra calculated from the dynamic frequency sweep together with shear creep and recovery curves quantify these assumptions. Dynamic rheology and shear creep and creep-recovery measurements are sensitive tools to get insights into the dispersion state of nanoparticles in the polymer composites, as well as the interactions (i.e., particle-polymer interaction, particle–particle interaction) in such systems.

Keywords Isotactic polypropylene (iPP) · Nano-zinc oxide (nano-ZnO) · Dynamic rheology · Shear creep and recovery · Dispersion · Interaction

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Introduction

Polymer nanocomposites have attracted great attention in industry and academy during the past decades owing to their unique properties [1–7]. Nanoparticles that dispersed on nanoscale could influence the properties of the resulting materials more prominently in contrast to microparticles with the same volume fraction. Particularly, incorporating rigid inorganic nanoparticles has been proved to be an effective and accessible method to improve both stiffness and toughness of general thermoplastics simultaneously [8, 9]. However, the good dispersion is a challenge since nanoparticles in molten polymers tend to form agglomerates resulting from the pronounced particle–particle interaction and high specific surface area. Moreover, poor affinity between nanoparticles and polymer matrix with different polarity also brings about unfavorable interfacial adhesion, followed by deteriorated properties of the resulting composites. With the addition of compatibilizer or surface modification treatment on the nanoparticles, the interfacial adhesion between the filler and the matrix could be enhanced, and unavoidable aggregation could be alleviated.

Investigating the dispersion of the nanoparticles and the interactions in the polymer composites is of great significance for better understanding the relationship between structure and properties and controlling the macro-performance of polymer nanocomposites. Scanning electron microscopy (SEM) is the most common method for assessing the quality of dispersion of nanoparticles in polymer matrix. In addition to qualitative observation, it could be used together with computer image analysis software for quantitative characterization. However, only particles that are completely or partially exposed on the observation surface could be detected. Moreover, the results are susceptible to the impurities and surface smoothness and flatness of the sample, followed by poor image contrast. In contrast to SEM, transmission electron microscope (TEM) test has the advantage of higher resolution. However, characterizing the dispersion of nanoparticles by TEM may lead to conclusions invalid, owing to the fact that only a tiny part of sample is examined by one micrograph [10]. Dynamic rheological measurement as a basic scientific approach has been considered to be an effective way to inspect the internal microstructure of polymer nanocomposites. Particularly, dynamic viscoelastic response in the low frequency region can provide important information related to the morphological structure of the system [11, 12]. This test is usually performed under a small amplitude (strain) condition, without destroying the structure of the material. Therefore, it is a powerful tool for assessing the state of dispersion of nanoparticles in the polymer composites [13, 14]. Creep is a time- and temperature-dependent phenomenon. Studies on the creep and recovery behavior of polymer nanocomposites are widely reported in the past few decades, mainly based on mechanical properties measurements. Since shear creep and recovery test could sensitively detect the changes in the microstructure of material, it is also a powerful means to assess the interactions in the polymer composites [15–19]. To the best of our knowledge, the influence of compatibilization on the morphological and structural properties of polymer nanocomposites, monitored by shear creep and recovery measurements, has been seldom reported in the open literature.

Isotactic polypropylene (iPP), a general thermoplastic, has been widely used in chemical, construction, packaging, automotive and other industrial applications owing to its excellent comprehensive performance. However, it is easy to degrade under effects of light, heat and oxygen [20, 21], which leads to a decrease in the performance of the product. Moreover, improving toughness is also one of the important topics for the high performance of iPP. Nano-zinc oxide (nano-ZnO), a multifunctional inorganic nanomaterial, has been the focus of numerous researchers because of its promising antistatic, antibacterial and ultraviolet light shield function [22–25]. Introducing a small amount of nano-ZnO can significantly improve the mechanical properties and aging resistance of iPP [26, 27]. Since iPP is hydrophobic, while nano-ZnO is hydrophilic, with characteristics of high surface energy and large specific surface area, the nanoparticles are difficult to disperse uniformly in the iPP matrix. In this paper, ZnO/iPP nanocomposites were prepared via melt blending, and the affinity of nano-ZnO to iPP was improved with the addition of maleic anhydride/styrene-grafted random copolypropylene (MPP). In addition to the observation of phase morphology by using SEM, dynamic rheological and creep and recovery tests in shear were used to further characterize the dispersion of ZnO nanoparticles and the interactions in the composites. Special attention was focused on the effect of nano-ZnO concentration and the introduction of MPP on the morphological and structural properties of nano-ZnO/iPP composites, which would help to fully understand the microstructure of composites and the mechanism of nanofillers on polymer matrix.

Experimental

Materials

Isotactic polypropylene (iPP, trade name T30S), with melt flow index of $3.2 \text{ g (10 min)}^{-1}$ (230 °C, 2.16 kg), was purchased from West Pacific Petrochemical Company (Dalian, China). Hydrophilic nano-zinc oxide (nano-ZnO) was supplied by Jing Rui New Material Co. Ltd. (Xuancheng, China) and had a density of 5.6 g cm^{-3} and a mean diameter of 30 nm. Maleic anhydride/styrene-grafted random copolypropylene (MPP) was synthesized in our laboratory [28], with melt flow index of $2.6 \text{ g (10 min)}^{-1}$ (230 °C, 2.16 kg) and MAH content of 2.1 wt.%. Powdery polypropylene random copolymer (PPR) with low melt flow index (0.5 g/10 min (230 °C, 2.16 kg)) was used as grafting matrix resin, which is beneficial to uniform mixing between grafting agents and powdery matrix resin. Moreover, styrene was used as a second grafting monomer, with the aim of inhibiting the degradation of iPP and improving the grafting degree. MPP was prepared in a twin-screw extruder. The premix of PPR, MAH, styrene (St), dicumyl peroxide (DCP) and antioxidant 1010 (terakis[methylene- β -(3,5-di-tert-butyl-4-hydroxyphenyl)-propionate] methane), which were mixed in a high-speed mixer in advance, was fed into the extruder under the required conditions. The processing temperatures from the first zone to the ninth zone were set as 170, 180, 190, 200, 200, 200, 200, 210 and 210 °C, respectively. The formulation of the sample is PPR 100 g, MAH 1.8 g, St 1.4 g, DCP 0.1 g and

antioxidant 1010 0.3 g. After the graft reaction was completed, MPP sample was dissolved in refluxing xylene at 140 °C, and excess acetone was then added to precipitate them. Acetone could precipitate only the grafted and ungrafted PPR. The precipitated samples were filtered, washed and dried under vacuum at 80 °C for 24 h.

Sample preparation

The nano-ZnO and iPP were melt blended by an internal mixer (PPt-3/LLZ-40, Guangzhou POTOP Experimental Analysis Instrument Co. Ltd, China) at 180 °C for 7 min, with a rotation speed of 60 rpm. The proportions of nano-ZnO in the composites were 0, 0.08, 0.16, 0.85, 1.77, 2.79, 3.90 vol.% (0, 0.5, 1, 5, 10, 15, 20% by weight), respectively. The concentration of MPP is 5 wt.% of the nano-ZnO content. Then the resulting composites were compression molded into disks in a press vulcanizer (406, Dongguan Xihua Testing Instrument Co. Ltd, China) with diameter of 25 mm and thickness of 1.2 mm at 10 MPa, 185 °C.

Characterization

Scanning electron microscopy (SEM)

A field emission scanning electron microscope (SUPRA 55, Zeiss, Germany) was used for the characterization of the cryofractured surface of the samples, with an accelerating voltage of 5 kV. The specimens were gold sputtered prior to SEM characterization.

Dynamic rheological properties

The dynamic rheological properties were measured using a rotational rheometer (Physica MCR 301, Anton-Paar, Austria). All the measurements were conducted in oscillatory shear mode with parallel plate geometry of diameter 25 mm and gap 1 mm under a nitrogen atmosphere. Frequency sweep tests were conducted within the linear viscoelastic region at 210 °C with an angular frequency range from 0.0398 to 628 rad s⁻¹.

Creep and recovery properties

Creep and recovery measurements were conducted in shear mode at 210 °C using a rotational rheometer (Physica MCR 301, Anton-Paar, Austria) in a nitrogen atmosphere, with parallel plate geometry of diameter 25 mm and gap 1 mm. The creep and recoverable strains were determined as a function of time ($t_{\text{creep}} = 10$ min and $t_{\text{recovery}} = 20$ min). The stress was fixed at 10 Pa, to ensure the measurements remained within the linear viscoelastic region.

Fourier transform infrared spectroscopy (FTIR)

Pure iPP, MPP and the ZnO/iPP composites before and after compatibilization were hot pressed into thin films at 180 °C for FTIR analysis, respectively. FTIR spectra were measured using a Fourier transform infrared spectrometer (Avatar 370, Nicolet, USA), which were recorded at the average of 32 scans in the standard wave number range of 500–4000 cm^{-1} at a resolution of 0.4 cm^{-1} .

Results and discussion

Phase morphology

The state of dispersion of nanoparticles has a significant influence on the final properties of the polymer nanocomposites. The final agglomerates size strongly depends on the concentration of nanoparticle. Figure 1 representatively presents the SEM micrographs of cryofractured surfaces of ZnO/iPP composites. It could be found that agglomeration becomes obvious with increasing nano-ZnO loading, as displayed in Fig. 1a, c and e. In order to enhance the interfacial interaction and promote the dispersion of nano-ZnO in the iPP matrix, maleic anhydride/styrene-grafted random copolypolypropylene (MPP) as a kind of macromolecular compatibilizer was introduced. The microscopic morphology of the cryofractured surface of ZnO/MPP/iPP composites is shown in Fig. 1d, and f. Comparing the SEM micrographs of the composites before and after compatibilization, it can be seen that the incorporating of MPP brings about the reduction of the number of big agglomerates of nano-ZnO.

Figure 2 presents the FTIR spectra of pure iPP, MPP, as well as the ZnO/iPP composites before and after compatibilization. In the case of MPP, besides the characteristic absorptions of the PP skeleton, there are two new bands at 1857 and 1782 cm^{-1} , which can be ascribed to anhydride groups, implying that maleic anhydride was grafted onto the PP backbones. A careful examination reveals that the characteristic peaks of the ZnO/iPP composites before and after compatibilization are almost the same, and there are no absorption peaks at 1857 and 1782 cm^{-1} . On the one hand, the MPP concentration in the composite is low. On the other hand, the maleic anhydride group in MPP can react with the hydroxyl groups on the surface of the nano-ZnO. In addition, PP macromolecular chains of MPP have good miscibility with the iPP matrix. Consequently, MPP could act as a bridge between the nanofillers and polymer matrix, improving the compatibility between the two phases and promoting the dispersion of nano-ZnO particles in the iPP.

Dynamic rheological behavior

As a method for characterizing the viscoelasticity of polymer, dynamic rheological measurement is also widely used to investigate the dispersion of fillers in polymer composites. Figure 3 displays the complex viscosity η^* of the ZnO/

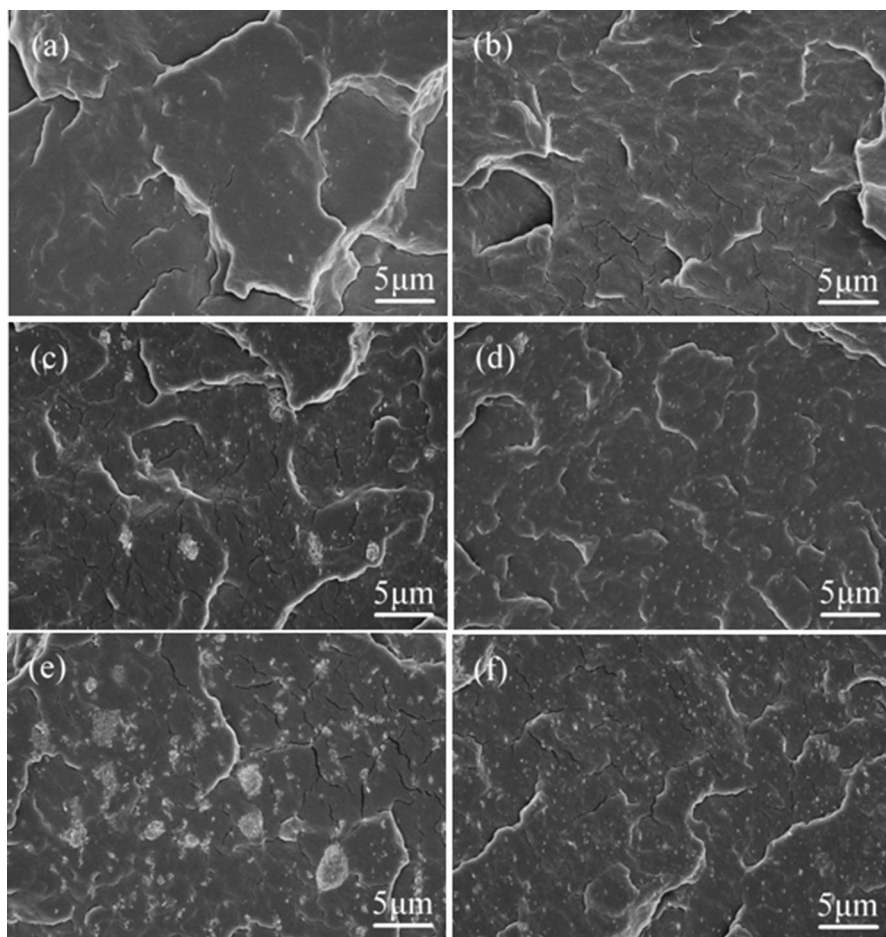


Fig. 1 SEM micrographs of cryofractured surfaces of ZnO/iPP composites before and after compatibilization **a** ZnO-0.16%/iPP, **b** ZnO-0.16%/MPP/iPP, **c** ZnO-0.85%/iPP, **d** ZnO-0.85%/MPP/iPP, **e** ZnO-2.79%/iPP, **f** ZnO-2.79%/MPP/iPP

iPP and ZnO/MPP/iPP composite melts as a function of angular frequency ω , as well as the comparison in η^* values of two systems at an angular frequency of 0.1 rad s^{-1} . It is apparent that the rheological behavior is more sensitive to the nano-ZnO concentration at low frequencies. The complex viscosity decreases remarkably with increasing angular frequency, especially for the high-loaded system. This could be attributed to the destruction of the percolation network and the decrease in the density of iPP chain entanglement, as well as the orientation of iPP chains in the shear field [29]. As displayed in Fig. 3c, the η^* values of ZnO/MPP/iPP composites are lower than those of ZnO/iPP composites with the same nano-ZnO loading, indicating that MPP inhibits the agglomeration of nano-ZnO

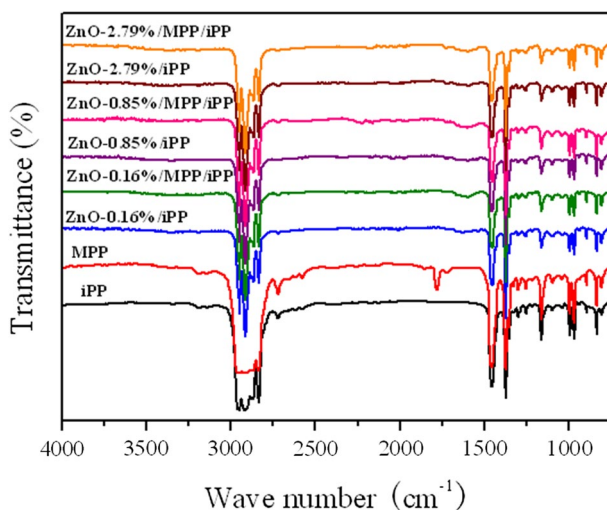


Fig. 2 FTIR spectra of pure iPP, MPP, as well as the ZnO/iPP composites before and after compatibilization

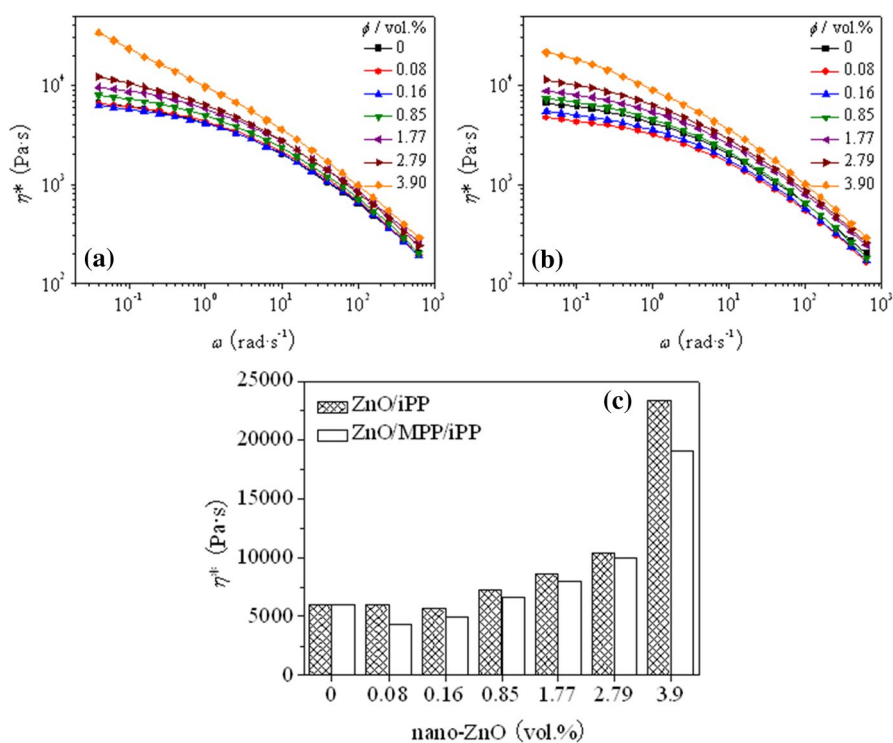


Fig. 3 Complex viscosity η^* of the ZnO/iPP (a) and ZnO/MPP/iPP (b) composite melts as a function of angular frequency ω , as well as the comparison in η^* values between two systems at an angular frequency of 0.1 rad s^{-1}

and promotes the dispersion of nano-ZnO particles in the iPP matrix. This is consistent with the results from SEM measurements.

At low loading at which no percolation network exists, the particle-polymer interaction is the main reason for the changes in rheological properties of the polymer composites. Due to poor adhesion between hydrophilic nano-ZnO and hydrophobic iPP, the iPP chains could slip on the surface of nano-ZnO particles, followed by the damage of chain entanglement network. With the increase in the nano-ZnO loading, the slippable area in the ZnO/iPP system may become larger. However, when the concentration of nano-ZnO is up to 0.85 vol.%, particle–particle interaction occurs through van der Waals forces, resulting in the agglomeration of nanoparticles (as shown in Fig. 1c). This prominently restricts the chain mobility of iPP. Therefore, the η^* value of the ZnO/iPP composite melts first decreases and then increases with increasing nano-ZnO loading. For the ZnO/MPP/iPP composites, the complex viscosity falls off obviously as the concentration of nano-ZnO lowers than 0.16 vol.%. It could be speculated that the interfacial lubrication may be greater than the interfacial adhesion. However, when the concentration is higher than 0.16 vol.%, opposite trend could be observed. In spite of the fact that MPP could promote the dispersion of nano-ZnO particles, agglomeration of nanoparticles was still unavoidable, especially at high concentration (as shown in Fig. 1f). Consequently, an increase in the complex viscosity with further increase in nano-ZnO loading could be observed.

Figure 4 presents the storage modulus G' of the ZnO/iPP and ZnO/MPP/iPP composite melts with respect to angular frequency ω . Note that the storage modulus of each system tends to be uniform in the high frequency region, implying that the viscoelasticity in the high frequency region is mainly determined by the polymer matrix. For the ZnO/iPP composites (Fig. 4a), when the concentration of nano-ZnO reaches 3.90 vol.%, the relationship between storage modulus and angular frequency deviates significantly from the classical linear viscoelastic theory [30], showing a special viscoelastic response with platform characteristics and percolation [31]. In the case of the ZnO/MPP/iPP composites (Fig. 4b), there is no obvious storage modulus platform with increasing nano-ZnO loading. This

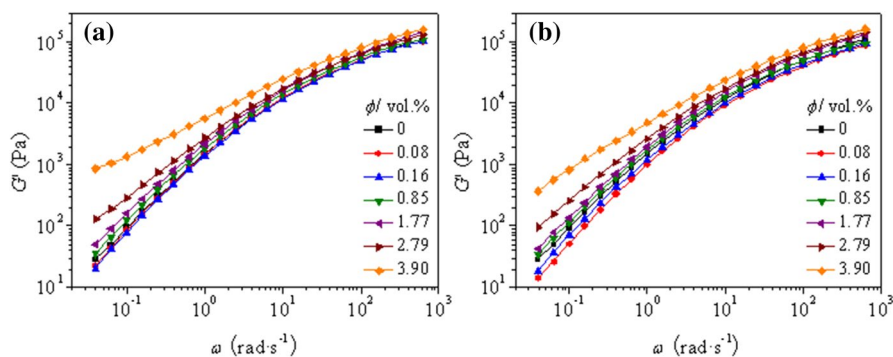


Fig. 4 Storage modulus G' of the ZnO/iPP (a) and ZnO/MPP/iPP (b) composite melts with respect to angular frequency ω

result further indicates that the addition of macromolecular compatibilizer can improve the dispersion of nano-ZnO particles in the iPP matrix.

Shear creep and recovery behavior

Creep, as a time dependent phenomenon, is of great importance for the applications requiring long-term durability and reliability. Polymer nanocomposites with excellent creep resistance have gained enormous interests for applications. Figure 5 shows the shear creep and recovery curves of ZnO/iPP composite melts before and after compatibilization. Interestingly, the creep and recovered strains of ZnO/iPP composite melts first increase and then decrease with respect to the nano-ZnO concentration (Fig. 5a). With the addition of 0.08 vol.% or 0.16 vol.% nano-ZnO, the strain increases significantly, mainly due to the slippage of iPP chains on the surface of the nano-ZnO particles. However, as the concentration of nano-ZnO reaches 0.85 vol.%, the strain begins to decrease. This indicates that nano-ZnO can only improve the creep resistance of iPP with a certain agglomeration size. With further increase in the nano-ZnO loading, the strain of the composite decreases significantly, implying that the percolation network formed by nano-ZnO particles leads to the enhancement of creep resistance of ZnO/iPP composites. For the ZnO/MPP/iPP composites, the variation of creep and recovered strains with respect to the nano-ZnO concentration shows the same trend as that of the ZnO/iPP composites, as displayed in Fig. 5b. It is worth noting that there are different responses of the recovery property of the composites with different nano-ZnO loading. The strain recovery ratio of high-loaded composites is much higher than that of low-loaded ones.

Comparing the shear creep and recovery curves of the composite melts with the same nano-ZnO loading, it could be found that when the concentration is 0.08, 0.16, 0.85 vol.%, the creep and recovered strains of ZnO/MPP/iPP composite melts are lower than those of the ZnO/iPP systems (Fig. 5c–e). However, for the composites with higher nano-ZnO concentration (1.77, 2.79, and 3.90 vol.%), opposite trend could be observed, as shown in Fig. 5f–h. Since the nano-ZnO loading is the same, it could be speculated that different particle–polymer or particle–particle network formed in the compatibilized composites in comparison with uncompatibilized ones, to be held responsible for this.

To clarify the different network formed in the composites, the relaxation behavior of the ZnO/iPP composites before and after compatibilization was characterized. It is normally thought the actual validity range of the weighted relaxation spectrum obtained from small amplitude oscillation tests is narrower than the experimental frequency limits [32]. Creep measurement is an effective method to detect the longer time portions of the relaxation spectrum to compensate for the limitation of dynamic frequency sweep experiment. According to the method proposed by Kaschta et al. [33–35], the dynamic viscoelastic parameters in the low frequency region could be obtained from creep data. By replacing the storage modulus curve in lower frequency region with the creep results, an enlarged storage modulus curve with wider frequency region could be obtained, as presented

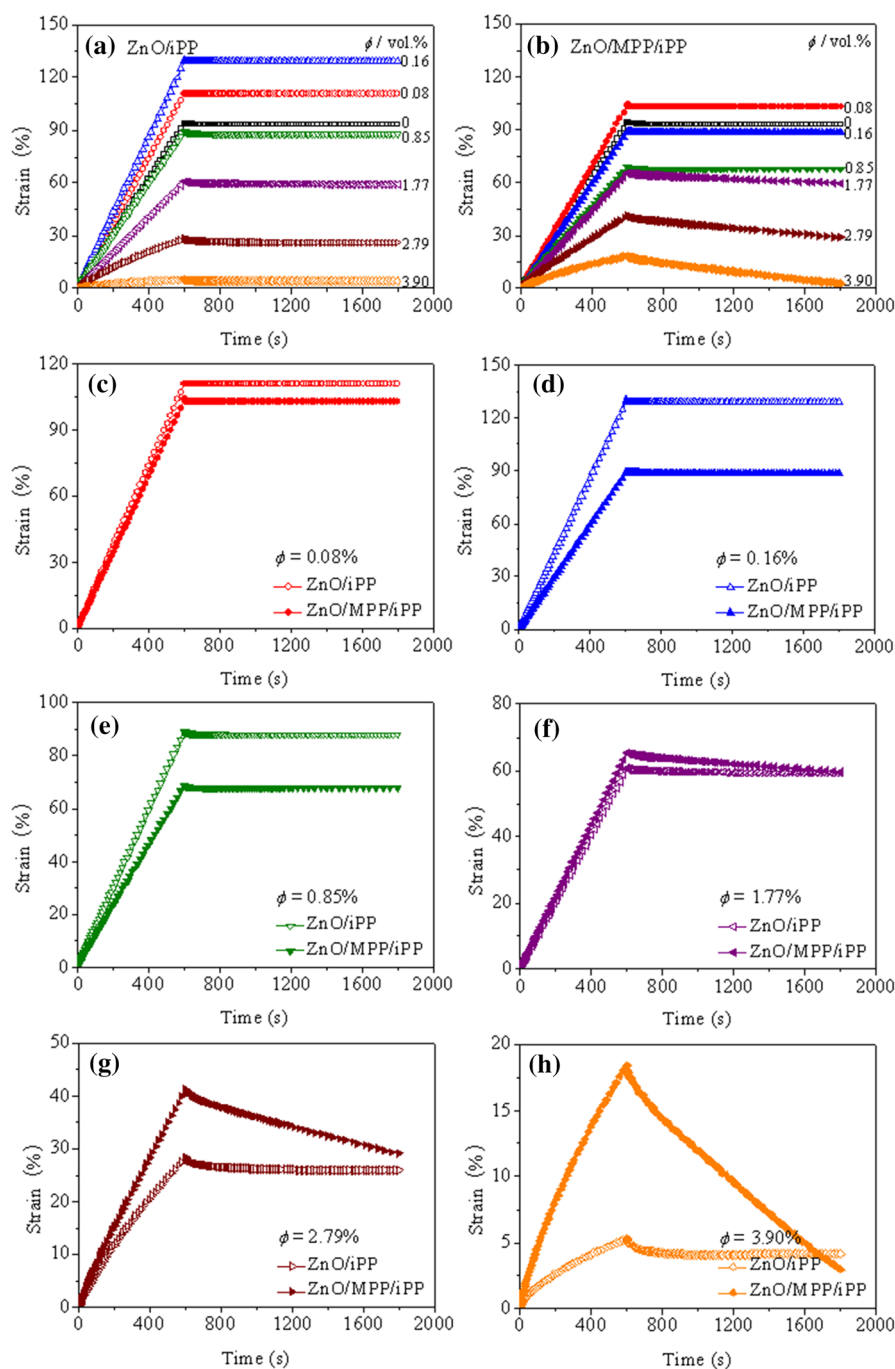


Fig. 5 Creep and recovery curves of nano-ZnO/iPP composite melts before and after compatibilization (a ZnO/iPP, b ZnO/MPP/iPP, c–h Comparison of creep and recovery curves at the same volume fraction, c $\phi = 0.08\%$, d $\phi = 0.16\%$, e $\phi = 0.85\%$, f $\phi = 1.77\%$, g $\phi = 2.79\%$, h $\phi = 3.90\%$)

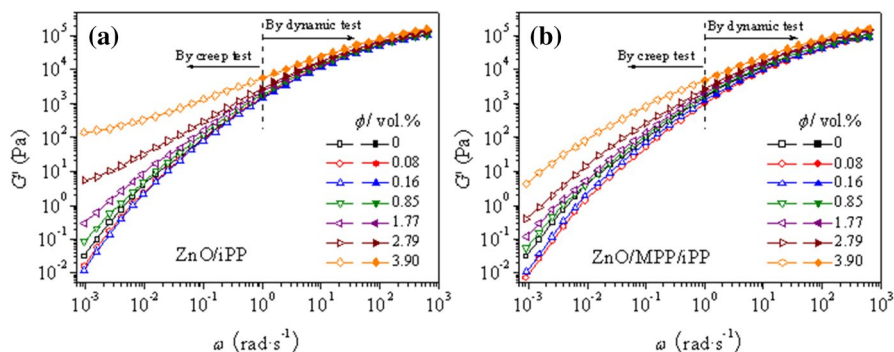


Fig. 6 Storage modulus G' from creep test (open symbols) and dynamic frequency sweep test (solid symbols) with respect to angular frequency ω for the ZnO/iPP (a) and ZnO/MPP/iPP (b) composite melts

in Fig. 6. It could be found that the data connect well. The storage modulus of the ZnO/iPP system keeps higher than that of ZnO/MPP/iPP system in the whole angular frequency region.

According to the method given by Tian et al. [36], the weighted relaxation spectrum could be calculated completely from Fig. 6 according to Eq. (1):

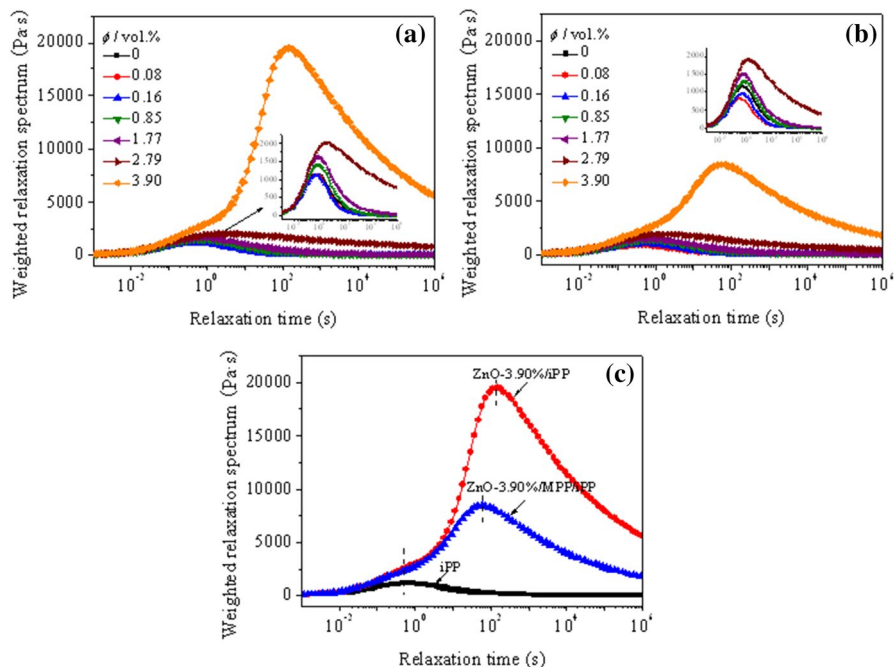


Fig. 7 Weighted relaxation spectra of ZnO/iPP composites before and after compatibilization (a ZnO/iPP, b ZnO/MPP/iPP, c Comparison of weighted relaxation spectra with 3.90 vol.% nano-ZnO)

$$G'(\omega) = \int_{-\infty}^{+\infty} H(\lambda) \omega^2 \lambda^2 / (1 + \omega^2 \lambda^2) d \ln \lambda \quad (1)$$

where λ expresses the relaxation time and $H(\lambda)$ represents the relaxation time spectrum. It can be seen from Fig. 7a that the relaxation time of pure iPP is 0.5 s. With incorporating nano-ZnO, the intensity of the weighted relaxation spectrum goes down slightly at first and then rises. When the concentration of nano-ZnO is 3.90 vol.%, the intensity increases significantly. The weighted relaxation spectrum of ZnO/MPP/iPP system shows a similar trend with respect to the nano-ZnO loading. However, the intensity of weighted relaxation spectrum for the composite with the same nano-ZnO loading falls off obviously, as presented in Fig. 7b. With incorporating 3.90 vol.% nano-ZnO, both the composites before and after compatibilization exhibit two relaxation time, which could be seen more clearly in Fig. 7c. Note that the shorter one corresponds to the relaxation of pure iPP. Based on the results of the relaxation time of polymer chain and nanofiller network reported in reference [37], it could be deduced that particle–polymer interaction results in an additional relaxation process with a long characteristic relaxation time, which is separated from the entanglement relaxation of the polymer. At lower filler loading at which no percolation network exists, the particle–polymer interaction is the main reason for the change in relaxation behavior of polymer nanocomposites. However, the particle–particle interaction and percolation network should contribute most to the relaxation properties at higher filler loading. With the addition of compatibilizer, the chain mobility mainly depends on the disentanglement between the PP chains in MPP and the iPP matrix, and the disentanglement time (56.2 s) is significantly shorter than the relaxation time of the iPP chains attached to the particle surface (141.3 s).

Combined with the previous discussion, we can deduce that MPP forms a stable interfacial layer around the nano-ZnO particles that facilitates the connection of adjacent nano-ZnO particles when incorporating high nano-ZnO loading. Physical crosslinking network formed by nano-ZnO particles supported MPP is prone to elastic deformation when subjected to shear stress, and the recovery is remarkable after the shear stress is removed. Therefore, worse creep resistance and better recovery properties for high-loaded composites after compatibilization in comparison with low-loaded ones could be observed.

Based on above results and discussion, we draw a schematic diagram of the dispersion and interaction mechanism of nano-ZnO in iPP matrix before and after compatibilization, as presented in Fig. 8. For the composites at nano-ZnO concentration below the percolation threshold, the dispersion of nano-ZnO in the iPP matrix is relatively uniform although some particles begin to agglomerate due to high surface energy and large specific surface area of nano-ZnO, as well as poor affinity of hydrophilic nano-ZnO to hydrophobic iPP (Fig. 8a–I). The incorporation of MPP improves the dispersion of nano-ZnO, followed by obviously reduced agglomerates (Fig. 8b–I). With increasing nano-ZnO loading, agglomeration of nano-ZnO particles in both ZnO/iPP and ZnO/MPP/iPP composites becomes serious. However, the main difference is that the nano-ZnO agglomerates are larger in the uncompatibilized samples

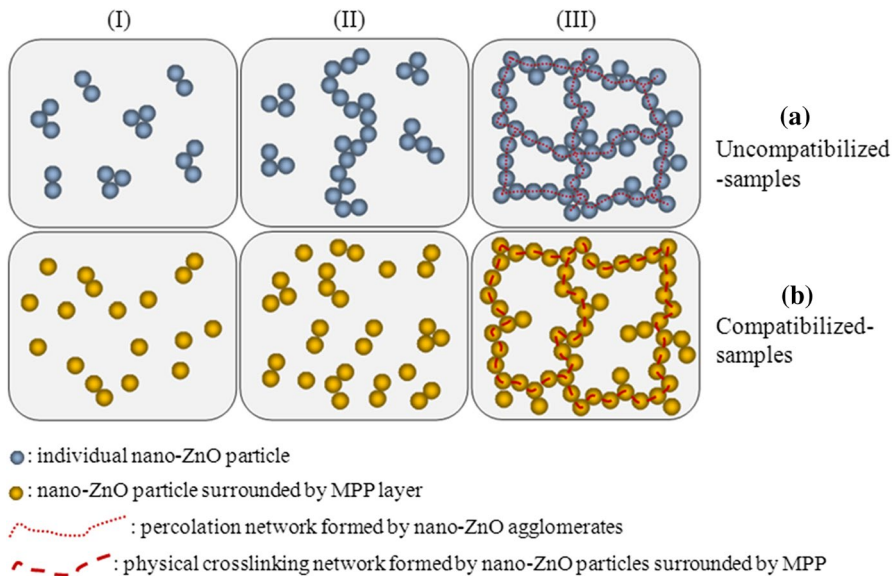


Fig. 8 Schematic diagram of the dispersion and interaction mechanism of nano-ZnO in iPP matrix before and after compatibilization

in comparison with compatibilized-samples, as shown in Fig. 8a-II and Fig. 8b-II, respectively. With the addition of 3.9 vol.% nano-ZnO, a percolation network is formed (Fig. 8a-III). The agglomeration of nano-ZnO particles will influence the contribution of particle–polymer interaction to the viscoelastic response. Large nano-ZnO agglomerates or more complete percolation network formation enhances the creep resistance of iPP. MPP forms a stable interfacial layer around the nano-ZnO particles that facilitates the connection of adjacent nano-ZnO particles when incorporating high nano-ZnO loading (Fig. 8b-III). Physical crosslinking network formed by nano-ZnO particles supported MPP is prone to elastic deformation when subjected to shear stress, and the recovery is prominent after the shear stress is removed.

Conclusions

In summary, dynamical rheology and shear creep and creep-recovery measurements were carried out on isotactic polypropylene (iPP) filled with different concentrations of nano-zinc oxide (nano-ZnO). The complex viscosity and storage modulus of the composite melts first decrease and then increase with increasing nano-ZnO loading. The pseudo-solid-like behavior appearing in high-loaded composites could be ascribed to the formation of the filler percolation network resulting from particle–particle interactions. The incorporating of maleic anhydride/styrene-grafted random copolypropylene (MPP) brings about the reduction of the number of big agglomerates of nano-ZnO, followed by the decrease in modulus and viscosity.

The agglomeration of nano-ZnO particles also influences the contribution of particle–polymer interaction to the viscoelastic response. Large nano-ZnO agglomerates or more complete percolation network formation enhances the creep resistance of iPP. MPP forms a stable interfacial layer around the nano-ZnO particles that facilitates the connection of adjacent nano-ZnO particles when incorporating high nano-ZnO loading. Physical crosslinking network formed by nano-ZnO particles supported MPP is prone to elastic deformation when subjected to shear stress, and the recovery is prominent after the shear stress is removed.

Dynamic rheological test could be used for the determination of the dispersion state of nanoparticles in the polymer composites, and shear creep and recovery measurement is a suitable method to reflect the interactions between the nano-filler and polymer matrix in the molten state. The combination of both facilitates a better understanding of the morphological and structural properties of polymer nanocomposites.

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